CSP-203: Software Tools & Technologies Lab

Chapter – 3

Document Processing: LaTeX (Lecture – 2)

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Typesetting Mathematics: Dollar Signs

- Why are dollar signs $ special? We use them to mark mathematics in text.

```
% not so good:
Let a and b be distinct positive
integers, and let c = a - b + 1.
% much better:
Let $a$ and $b$ be distinct positive
integers, and let $c = a - b + 1$.
```

Let a and b be distinct positive integers, and let c = a - b + 1.

Let *a* and *b* be distinct positive integers, and let *c* = *a* – *b* + 1.

- Always use dollar signs in pairs — one to begin the mathematics, and one to end it.
- LaTeX handles spacing automatically; it ignores your spaces.

```
Let $y=mx+b$ be \ldots

Let $y = m  x  +  b$ be \ldots
```

Let *y* = *mx* + *b* be . . .

Let *y* = *mx* + *b* be . . .

Typesetting Mathematics: Notation

- Use caret ^ for superscripts and underscore _ for subscripts.

```
$y = c_2 x^2 + c_1 x + c_0$
```

$y = c_2 x^2 + c_1 x + c_0$

- Use curly braces { } to group superscripts and subscripts.

```
$F_n = F_n-1 + F_n-2$      % oops!

$F_n = F_{n-1} + F_{n-2}$ % ok!
```

$F_n = F_n - 1 + F_n - 2$

$F_n = F_{n-1} + F_{n-2}$

- There are commands for Greek letters and common notation.

```
$\mu = A e^{Q/RT}$

$\Omega = \sum_{k=1}^{n} \omega_k$
```

$\mu = Ae^{Q/RT}$

$\Omega = \sum_{k=1}^{n} \omega_k$

Typesetting Mathematics: Displayed Equations

- If it's big and scary, *display* it on its own line using \begin{equation} and \end{equation}.

```
The roots of a quadratic equation are given by
\begin{equation}
x = \frac{-b \pm \sqrt{b^2 - 4ac}}
                    {2a}
\end{equation}
where $a$, $b$ and $c$ are \ldots
```

The roots of a quadratic equation are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (2)$$

where a, b and c are . . .

Caution: LaTeX mostly ignores your spaces in mathematics, but it can't handle blank lines in equations — don't put blank lines in your mathematics.

Interlude: Environments

- equation is an *environment* — a context.
- A command can produce different output in different contexts.

```
We can write
$ \Omega = \sum_{k=1}^{n} \omega_k $
in text, or we can write
\begin{equation}
   \Omega = \sum_{k=1}^{n} \omega_k
\end{equation}
to display it.
```

We can write $\Omega = \sum^{n}_{k=1} \omega_k$ in text, or we can write

$$\Omega = \sum_{k=1}^{n} \omega_k \quad (3)$$

to display it.

- Note how the Σ is bigger in the equation environment, and how the subscripts and superscripts change position, even though we used the same commands.

In fact, we could have written $...$ as \begin{math}...\end{math}.

Interlude: Environments

- The \begin and \end commands are used to create many different environments.
- The itemize and enumerate environments generate lists.

```
\begin{itemize} % for bullet points
\item Biscuits
\item Tea
\end{itemize}

\begin{enumerate} % for numbers
\item Biscuits
\item Tea
\end{enumerate}
```

- Biscuits
- Tea

1. Biscuits
2. Tea

Interlude: Packages

- All of the commands and environments we've used so far are built into LaTeX.
- *Packages* are libraries of extra commands and environments. There are thousands of freely available packages.
- We have to load each of the packages we want to use with a `\usepackage` command in the *preamble*.
- Example: amsmath from the American Mathematical Society.

```
\documentclass{article}
\usepackage{amsmath} % preamble
\begin{document}
% now we can use commands from amsmath here...
\end{document}
```

Typesetting Mathematics: Examples with amsmath

- Use equation* ("equation-star") for unnumbered equations.

```
\begin{equation*}
    \Omega = \sum_{k=1}^{n} \omega_k
\end{equation*}
```

$$\Omega = \sum_{k=1}^{n} \omega_k$$

- LaTeX treats adjacent letters as variables multiplied together, which is not always what you want. amsmath defines commands for many common mathematical operators.

```
\begin{equation*} % bad!
  min_{x,y} (1-x)^2 + 100(y-x^2)^2
\end{equation*}
\begin{equation*} % good!
\min_{x,y}{(1-x)^2 + 100(y-x^2)^2}
\end{equation*}
```

$$min_{x,y}(1-x)^2+100(y-x^2)^2$$

$$\min_{x,y} (1-x)^2 + 100(y-x^2)^2$$

- You can use \operatorname for others.

```
\begin{equation*}
\beta_i =
\frac{\operatorname{Cov}(R_i, R_m)}
        {\operatorname{Var}(R_m)}
\end{equation*}
```

$$\beta_i = \frac{\operatorname{Cov}(R_i, R_m)}{\operatorname{Var}(R_m)}$$

Typesetting Mathematics: Examples with amsmath

- Align a sequence of equations at the equals sign

$$\begin{aligned}(x+1)^3 &= (x+1)(x+1)(x+1)\\ &= (x+1)(x^2+2x+1)\\ &= x^3+3x^2+3x+1\end{aligned}$$

with the align* environment.

```
\begin{align*}
   (x+1)^3  &=  (x+1)(x+1)(x+1)  \\
            &= (x+1)(x^2 + 2x + 1) \\
            &= x^3 + 3x^2 + 3x + 1
\end{align*}
```

- An ampersand & separates the left column (before the =) from the right column (after the =).
- A double backslash \\ starts a new line.

Typesetting Exercise 2

Typeset this in LaTeX:

Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $N(0, \sigma^2)$.

Click to open this exercise in **Overleaf**

- Hint: the command for ∞ is `\infty`.
- Once you've tried, click here to see my solution.

End of Part 1

- Congrats! You've already learned how to . . .
  - Typeset text in LaTeX.
  - Use lots of different commands.
  - Handle errors when they arise.
  - Typeset some beautiful mathematics.
  - Use several different environments.
  - Load packages.
- That's amazing!
- In Part 2, we'll see how to use LaTeX to write structured documents with sections, cross references, figures, tables and bibliographies. See you then!

Latex Outline

- Tables
- Figures
- Algorithms
- Bibliography
- Overleaf Demo
- Putting it all together

Tables

\begin{tabular}[pos]{table spec}

table spec	l	left-justified column
	c	centered column
	r	right-justified column
	p{'width'}	paragraph column with text vertically aligned at the top
	m{'width'}	paragraph column with text vertically aligned in the middle (requires array package)
	b{'width'}	paragraph column with text vertically aligned at the bottom (requires array package)
	|	vertical line
	||	double vertical line
pos	b	bottom
	c	center (default)
	t	top

Tabular

- Used in LaTeX environment to create tables using \begin{tabular} and \end{tabular} commands

```
\begin{tabular}{c c c}
    ......................
    ......................
\end{tabular}
```

- **Basic example of using tabular**

```
\documentclass{article}
\begin{document}
\begin{center}
\begin{tabular}{ c c c }
 cell1 & cell2 & cell3 \\
 cell4 & cell5 & cell6 \\
 cell7 & cell8 & cell9
\end{tabular}
\end{center}
\end{document}
```

Output:

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Tabular (1)

- **Add lines between columns and rows in a table**

```
\documentclass{article}
\begin{document}
\begin{center}
\begin{tabular}{ |c|c|c| }
 \hline
 cell1 & cell2 & cell3 \\ \hline
 cell4 & cell5 & cell6 \\ \hline
 cell7 & cell8 & cell9 \\  \hline
\end{tabular}
\end{center}
\end{document}
```

Output:

cell1	cell2	cell3
cell4	cell5	cell6
cell7	cell8	cell9

Latex: Figures

- Used in LaTeX environment to add figures using \begin{figure} and \end{figure} commands

```
\begin{figure}[h]
\centering
 \includegraphics{Image-Name}
 \caption{Figure description}
\end{figure}
```

- **Basic example of using figure**

```
\documentclass{article}
\usepackage{graphicx}  % Package for adding figures
\begin{document}
\begin{figure}[h]
\centering
\includegraphics{galaxy.jpg}
 \caption{Galaxy of Stars}
\end{figure}
\end{document}
```

- **Output:**



Figure 1: Galaxy of Stars

Note the size of the image can be changed using [scale= value] after includegraphics

LATEX: algorithm

- To typeset algorithms or pseudocode in LaTeX you can use one of the following options:
  - algpseudocode; algcompatible; and algorithmic packages to typeset algorithm bodies
  - algorithm package for captioning the algorithm.
  - algorithm2e package.

```
\document class{article}
\usepackage{algpseudocode}
\begin{document}
\begin{algorithmic}
\State $i \gets 10$
\If{$i\geq 5$}
   \State $i \gets i-1$
\Else
   \If{$i\leq 3$}
      \State $i \gets i+2$
   \EndIf
\EndIf
\end{algorithmic}
\end{document}
```

Output;

$i \leftarrow 10$
if $i \geq 5$ **then**
  $i \leftarrow i - 1$
else
  if $i \leq 3$ **then**
    $i \leftarrow i + 2$
  end if
end if

Note:
Need not to load the algorithm2e, algcompatible, algorithmic packages if you have already loaded algpseudocode.

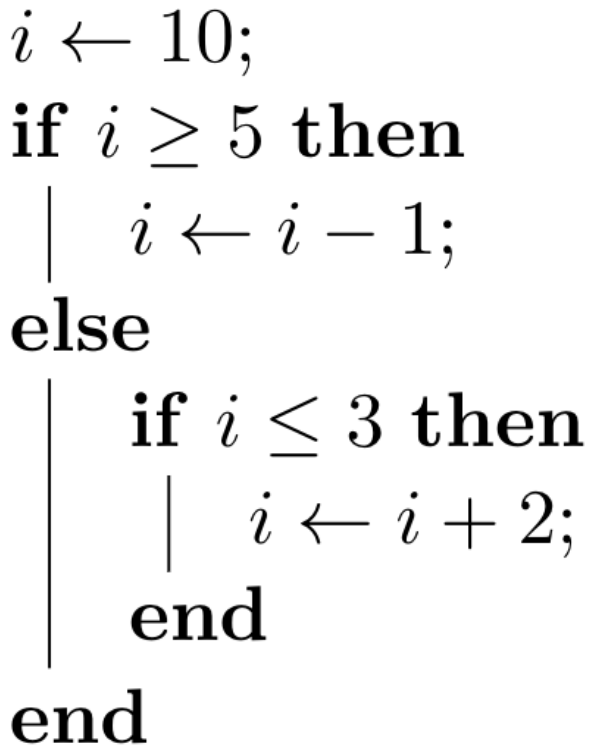
LATEX : algorithm

- To typeset algorithms or pseudocode in LaTeX you can use one of the following options:
  - With the algorithm2e

hello.tex

```
\documentclass{article}
\usepackage{algorithm2e}
\begin{document}
\begin{algorithm}
$i \gets 10$\;
\eIf{$i\geq 5$}
{
   $i \gets i-1$\;
}{
   \If{$i \leq 3$}
   {
      $i \gets i+2$\;
   }
}
\end{algorithm}
\end{document}
```

output



LATEX : Bibliography

```
\begin{thebibliography}{1}
\bibitem{3gpp300}
3GPP TS 38.300, "NR and NG-RAN Overall Description; Stage 2 (Release 15)", V15.1.0 (2018-03).
\bibitem {3gpp801}
3GPP TR 38.801, ``Study on new radio access technology: Radio access architecture and interfaces (Release 14)",
V14.0.0 (2017-03).
\bibitem {ngfi}
``Next generation fronthaul interface (1914) working group," http://sites.ieee.org/sagroups-1914 (accessed 04/2018).
\end{thebibliography}
```

References

1. 3GPP TS 38.300, "NR and NG-RAN Overall Description; Stage 2 (Release 15)", V15.1.0 (2018-03).
2. 3GPP TR 38.801, "Study on new radio access technology: Radio access architecture and interfaces (Release 14)", V14.0.0 (2017-03).
3. "Next generation fronthaul interface (1914) working group," http://sites.ieee.org/sagroups-1914 (accessed 04/2018).

LATEX : Bibliography

```
\bibliography{bibilo}
\bibliographystyle{IEEEtran}
```

```
@inproceedings{ujjayant2025,
  title={{TCP-Cubiwood: Enhanced Cubic Congestion Control for mmWave and Future Networks}},
  author={Ujjayant Prakash; Anand M. Baswade},
  booktitle={COMSNETS 2025, 17th International Conference on COMmunication Systems and NETworks
(COMSNETS)},
  year={2025}
}
```

```
@misc{iperf3,  title={{iperf3 Download Page}},
author={{iperf3}},
year={{n.d.}},
url={https://iperf.fr/iperf-download.php},
note={{Accessed: Date}}}
```

iperf3, "iperf3 Download Page," n.d., Accessed: Date. [Online].
Available: https://iperf.fr/iperf-download.php

An Interactive Introduction to LaTeX

Part 2: Structured Documents & More

Dr John D. Lees-Miller





Outline

- Structured Documents
  - Title and Abstract
  - Sections
  - Labels and Cross-References
  - Exercise
- Figures and Tables
  - Graphics
  - Floats
  - Tables
- Bibliographies
  - bibTEX
  - Exercise
- What's Next?
  - More Neat Things
  - More Neat Packages
  - Installing LATEX
  - Online Resources

Structured Documents

- In Part 1, we learned about commands and environments for typesetting text and mathematics.
- Now, we'll learn about commands and environments for structuring documents.
- You can try out the new commands in Overleaf:

Click here to open the example document in **Overleaf**

For best results, please use Google Chrome or a recent FireFox.

- Let's get started!

Title and Abstract

- Tell LaTeX the \title and \author names in the preamble.
- Then use \maketitle in the document to actually create the title.
- Use the abstract environment to make an abstract.

```
\documentclass{article}

\title{The Title}

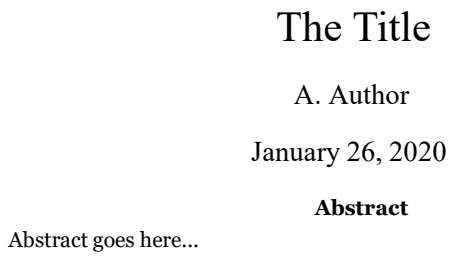
\author{A. Author}

\date{\today}

\begin{document}
\maketitle

\begin{abstract} Abstract goes here...
\end{abstract}

\end{document}
```



Sections

- Just use \section and \subsection.
- Can you guess what \section* and \subsection* do?

```
\documentclass{article}
\begin{document}

\section{Introduction} The problem of

\ldots


\section{Method}

We investigate \ldots

\subsection{Sample  Preparation}

\subsection{Data  Collection}

\section{Results}

\section{Conclusion}

\end{document}
```

1 Introduction

The problem of . . .

2 Method

We investigate . . .

2.1 Sample Preparation

2.2 Data Collection

3 Results

4 Conclusion

Labels and Cross-References

- Use \label and \ref for automatic numbering.
- The amsmath package provides \eqref for referencing equations.

```
\documentclass{article}
\usepackage{amsmath} % for \eqref
\begin{document}

\section{Introduction}
\label{sec:intro}
In Section \ref{sec:method}, we \ldots

\section{Method}
\label{sec:method}

\begin{equation}
\label{eq:euler} e^{i\pi} + 1 = 0
\end{equation}

By \eqref{eq:euler}, we have \ldots

\end{document}
```

1 Introduction

In Section 2, we . . .

2 Method

$$e^{i\pi} + 1 = 0 \quad (1)$$

By (1), we have . . .

Structured Documents Exercise

Typeset this short paper in LaTeX: [1]

Click to open the paper

Make your paper look like this one. Use \ref and \eqref to avoid explicitly writing section and equation numbers into the text.

Click to open this exercise in **Overleaf**

- Once you've tried, click here to see my solution.

Graphics

- Requires the graphicx package, which provides the \includegraphics command.
- Supported graphics formats include JPEG, PNG and PDF (usually).

```
\includegraphics[
  width=0.5\textwidth]{gerbil}

\includegraphics[
  width=0.3\textwidth,
  angle=270]{gerbil}
```





Interlude: Optional Arguments

- We use square brackets `[ ]` for optional arguments, instead of braces `{ }`.
- `\includegraphics` accepts optional arguments that allow you to transform the image when it is included. For example, `width=0.3\textwidth` makes the image take up 30% of the width of the surrounding text (`\textwidth`).
- `\documentclass` accepts optional arguments, too. Example:

  ```
  \documentclass[12pt,twocolumn]{article}
  ```

  makes the text bigger (12pt) and puts it into two columns.
- Where do you find out about these? See the slides at the end of this presentation for links to more information.

Floats

- Allow LaTeX to decide where the figure will go (it can “float”).
- You can also give the figure a caption, which can be referenced with \ref.

```
\documentclass{article}
\usepackage{graphicx}
\begin{document}

Figure \ref{fig:gerbil} shows \ldots

\begin{figure}
\centering
\includegraphics[% width=0.5\textwidth]{gerbil}
\caption{\label{fig:gerbil}Aww\ldots.}
\end{figure}

\end{document}
```



Figure 1: Aww. . . .

Figure 1 shows . . .

Tables

- Tables in Latex take some getting used to.
- Use the tabular environment from the tabularx package.
- The argument specifies column alignment — **l**eft, **r**ight, **r**ight.

```
\begin{tabular}{lrr}
Item          & Qty & Unit \$ \\
Widget & 1    & 199.99  \\
Gadget &  2   & 399.99  \\
Cable  &   3  & 19.99   \\
\end{tabular}
```

Item	Qty	Unit $
Widget	1	199.99
Gadget	2	399.99
Cable	3	19.99

- It also specifies vertical lines; use \hline for horizontal lines.

```
\begin{tabular}{|l|r|r|} \hline
Item          & Qty & Unit \$   \\\hline
Widget & 1    & 199.99        \\
Gadget & 2    & 399.99        \\
Cable  & 3    & 19.99         \\\hline
\end{tabular}
```

Item	Qty	Unit $
Widget	1	199.99
Gadget	2	399.99
Cable	3	19.99

- Use an ampersand & to separate columns and a double backslash \\ to start a new row (like in the align* environment that we saw in part 1).

bibtex 1

- Put your references in a .bib file in ‘bibtex’ database format:

```
@Article{Jacobson1999Towards, author = {Van
    Jacobson},
    title = {Towards the Analysis of Massive Multiplayer Online
                    Role-Playing Games},
    journal = {Journal of Ubiquitous Information}, Month = jun,
    Year = 1999,
    Volume = 6,
    Pages = {75--83}}

@InProceedings{Brooks1997Methodology,
    author = {Fredrick P. Brooks and John Kubiatowicz and Christos Papadimitriou},
    title = {A Methodology for the Study of the
                    Location-Identity Split}, booktitle = {Proceedings of
    OOPSLA}, Month = jun,
    Year = 1997}
```

- Most reference managers can export to bibtex format.

bibTex 2

- Each entry in the .bib file has a *key* that you can use to reference it in the document. For example, Jacobson1999Towards is the key for this article:

```
@Article{Jacobson1999Towards, author = {Van
    Jacobson},
    ...
}
```

- It's a good idea to use a key based on the name, year and title.
- LATEX can automatically format your in-text citations and generate a list of references; it knows most standard styles, and you can design your own.

bibTex 3

- Use the natbib package[2] with \citet and \citep.
- Reference \bibliography at the end, and specify a \bibliographystyle.

```
\documentclass{article}
\usepackage{natbib}
\begin{document}

\citet{Brooks1997Methodology} show that
\ldots. Clearly, all odd numbers are prime
\citep{Jacobson1999Towards}.

\bibliography{bib-example}
% if `bib-example' is the name of
% your bib file

\bibliographystyle{plainnat}
% try changing to abbrvnat

\end{document}
```

Brooks et al. [1997] show that . . . . Clearly, all odd numbers are pri [Jacobson, 1999].

References

Fredrick P. Brooks, John Kubiatowicz, and Christos Papadimitriou. A meth ology for the study of the location-identity split. In *Proceedings of OOPSL* June 1997.

Van Jacobson. Towards the analysis of massive multiplayer online role-playi games. *Journal of Ubiquitous Information*, 6:75–83, June 1999.

[2]There is a new package with more features named biblatex but most of the articles templates still use natbib.

Exercise: Putting it All Together

Add an image and a bibliography to the paper from the previous exercise.

1. Download these example files to your computer.

   Click to download example image

   Click to download example bib file

2. Upload them to Overleaf (use the project menu).

More Neat Things

- Add the \tableofcontents command to generate a table of contents from the \section commands.
- Change the \documentclass to

  \documentclass{scrartcl}

  or

  \documentclass[12pt]{IEEEtran}
- Define your own command for a complicated equation:

```
\newcommand{\rperf}{%
\rho_{\text{perf}}}
$$
\rperf = {\bf c}'{\bf X} + \varepsilon
$$
```

$$\rho_{\text{perf}} = \mathbf{c}'\mathbf{X} + \varepsilon$$

More Neat Packages

- beamer: for presentations (like this one!)
- todonotes: comments and TODO management
- tikz: make amazing graphics
- pgfplots: create graphs in LaTeX
- listings: source code printer for LaTeX
- spreadtab: create spreadsheets in LaTeX
- gchords, guitar: guitar chords and tabulature
- cwpuzzle: crossword puzzles

See https://www.overleaf.com/latex/examples and http://texample.net for examples of (most of) these packages.

https://www.overleaf.com/learn/latex/Beamer

Installing Latex

- To run Latex on your own computer, you'll want to use a Latex distribution. A distribution includes a latex program and (typically) several thousand packages.

- On Windows: MikTEX or TEXLive
- On Linux: TEXLive
- On Mac: MacTEX

- You'll also want a text editor with Latex support. See http://en.wikipedia.org/wiki/Comparison_of_TeX_editors for a list of (many) options.
- You'll also have to know more about how latex and its related tools work — see the resources on the next slide.

Online Resources

- The Overleaf Learn Wiki — hosts these slides, more tutorials and reference material
- The LaTeX Wikibook — excellent tutorials and reference material.
- TEX Stack Exchange — ask questions and get excellent answers incredibly quickly
- LATEX Community — a large online forum
- Comprehensive TEX Archive Network (CTAN) — over four thousand packages plus documentation
- Google will usually get you to one of the above.

Thank you!

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